

RESEARCH ARTICLE

Interaction-Aware Constraint Interface for Path–Velocity Planning in Dynamic Multi-Obstacle Traffic

JiaWang Liao^a, YuFei Hu^a, ChaoYang Shi^a and ChengJiao Sun^a

^aSchool of Computer Engineering, Hubei University of Arts and Science, 296 Longzhong Road, Xiangyang 441053, Hubei, China

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ABSTRACT

In two-stage path–velocity planning, time-blind obstacle bounds and local passing-side decisions can rule out feasible passages and cause unnecessary stops in dense traffic. We present MIKU, a constraint-generation method that constructs lateral bands from estimated arrival times, selects a sequence of bands across road stations, and maps safe passing windows into bounds for the existing path and speed quadratic programs. For a fixed-section interval model, the widest lateral band can be identified by sorting and scanning obstacle intervals in $O(k \log k)$ time. In 3,500 paired structured-roadway simulations spanning seven traffic scenarios, MIKU increased collision-free goal reaching from 62.69% for an internal time-blind, fixed-side baseline to 77.46% (paired difference: 14.77 percentage points; 95% confidence interval: 13.54–15.97). In a separate set of 700 rolling-replanning simulations, success increased from 62.29% to 72.29%. Complete-call planning latency had a median of 9.98 ms and a 99th percentile of 221.80 ms in the paired benchmark. These findings show that interaction-aware constraint selection improves path–velocity planning under the tested simulation conditions.

KEYWORDS

autonomous driving; path–velocity decomposition; space–time homotopy; dynamic obstacles; motion planning

1. Introduction

In urban mixed-traffic autonomous driving, planners built on path–velocity decomposition emit unnecessary fallback stops because time-blind path bounds and obstacle-wise side choices discard interaction timing before either solver runs. For autonomous-vehicle system integrators and safety engineers, these artificial deadlocks directly degrade ride quality and route efficiency, creating a measurable gap between theoretical feasibility and operational performance in dense traffic. This paper develops MIKU, a constraint reconstruction framework for two-stage quadratic programming path–velocity decomposition architectures, evaluated on 4,000 deterministic interval instances and 3,500 paired simulated scenarios via collision-free goal reaching, progress ratio, jerk RMS, and planning-time quantiles, with degraded-stop rate directly reflecting the operational trade-offs among safety, progress, ride comfort, and real-time computation (González, Pérez, Milanés, & Nashashibi, 2016; Paden, Čáp, Yong, Yershov, & Frazzoli, 2016; Schwarting, Alonso-Mora, & Rus, 2018).

Specifically, path–velocity decomposition separates lateral path generation in Frenet station–lateral coordinates s – l from longitudinal speed optimization in station–time coordinates s – t (Kant & Zucker, 1986; Werling, Ziegler, Kammel, & Thrun, 2010; Zhou et al., 2021). Industrial implementations typically solve two structured convex quadratic programs that expose environmental constraints through spatial path boundaries and temporal ST boundaries (Fan et al.,

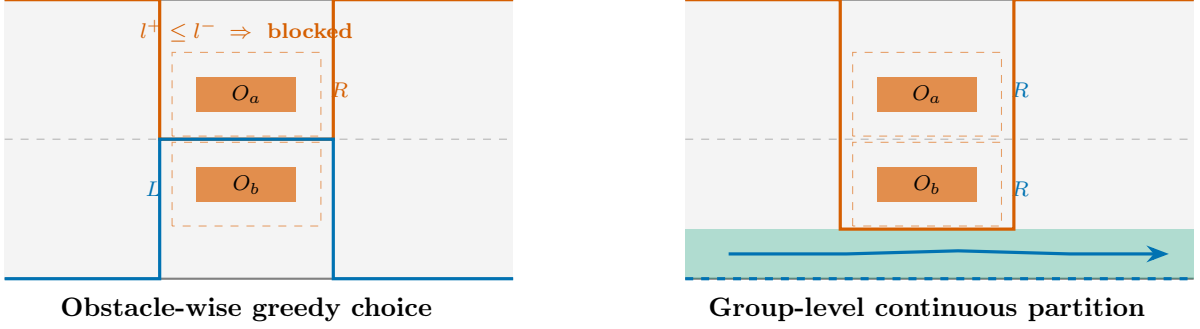


Figure 1. Greedy obstacle-wise side choices make $l^+(s)$ and $l^-(s)$ cross and trigger a fallback stop (a). MIKU’s group-level partition keeps a continuous passage for $\mathcal{T}_{ego}$ (b).

2018; Wang et al., 2022; Y. Zhang et al., 2020). In dense dynamic scenes this separation discards two kinds of information: a path boundary from a current snapshot or a horizon union cannot distinguish an area occupied now from one free at the ego’s arrival, and irreversible obstacle-wise side choices can raise the lower and lower the upper path bound in mutually incompatible ways even when a jointly selected passage remains open.

While joint space–time search, mixed-integer optimization, and spatio-temporal corridor methods coordinate spatial and temporal decisions simultaneously (Kessler, Esterle, & Knoll, 2023; McNaughton, Urmson, Dolan, & Lee, 2011; Yoon, Kim, Lee, Seo, & Yi, 2024), they typically expand state spaces or alter optimization structures, making them incompatible with production two-QP architectures. Consequently, current industrial PVD implementations suffer from three fundamental limitations: first, time-blind path boundaries merge occupancies across the prediction horizon, losing arrival-conditioned clearance before lateral optimization; second, greedy, obstacle-wise side choices cannot capture group-level continuous passages, causing lower and upper path bounds to cross and falsely signal a blockage even when viable corridors exist; and third, existing cross-stage coordination mechanisms cannot transfer spatio-temporal homotopy decisions without modifying downstream convex solvers. As illustrated in Figure 1, MIKU resolves these challenges directly at the constraint interface by constructing time-dependent lateral bands, selecting spatial and temporal homotopy classes, and mapping them into standard box constraints accepted by the two QPs.

The principal contributions of this work are as follows:

- (1) **Arrival-time-dependent spatial projection:** Addressing the first gap of time-blind occupancy merging, we formulate an arrival-time-dependent mapping that projects obstacle occupancies into center-coordinate path bounds based on estimated ego arrival times, ensuring the path planner considers only obstacles present when the vehicle traverses each station. This mechanism is developed in Section 4.4 and isolated by ablation A1 in Table 4.
- (2) **Group-level continuous lateral partitioning and fast scan:** Addressing the second gap of greedy obstacle-wise deadlocks, we prove the continuous-partition lemma and maximum-gap theorem in Section 4.3. These theoretical foundations reduce the exponential 2^k side assignments for k obstacles to an exact $O(k \log k)$ scan under the fixed-section interval model, verified against exhaustive search across 4,000 deterministic instances and evaluated by ablations A3–A4 in Table 4.
- (3) **Layered temporal homotopy graph and QP-preserving transfer:** Addressing the third gap of cross-stage constraint loss, we transfer spatial decisions to longitudinal planning via uncertainty-expanded occupancy tubes, safe passage windows, and a layered temporal homotopy graph in Section 4.5. Timing conditions are formulated as linear box constraints that preserve downstream QP structures and computational efficiency, with empirical verification through ablations A5–A6 in Table 4.
- (4) **Comprehensive empirical and system-level validation:** We systematically evaluate MIKU across 3,500 paired scenarios with rigorous confidence intervals and ablations, validate

closed-loop execution in 700 rolling-replanning episodes, and demonstrate software-interface compatibility with the production Apollo 11.0 autonomous driving system in Sections 5 and 6.

Section 2 positions the interface among joint and decomposed planners; Section 3 defines the PVD interface and its failure mechanisms; Section 4 presents MIKU; Sections 5–6 cover evaluation and results; Sections 7–9 discuss and conclude.

2. Related work

2.1. Joint space–time planning

Joint planners retain space–time coupling via an expanded state. Conformal spatiotemporal lattices coordinate position, velocity, and time (McNaughton et al., 2011); instantaneous-analysis and SLT formulations couple lateral and longitudinal decisions in dense traffic (Hu et al., 2023, 2025; Qiao, 2025; X. Yang & Li, 2021); mixed-integer formulations encode side choices as discrete variables (Kessler et al., 2023). These differ structurally from MIKU’s sequential convex path and piecewise-jerk speed optimizations; our finite-grid joint search is a computational reference on the same collision checker.

2.2. Decomposed and iterative coordination

PVD planners optimize path then speed (Kant & Zucker, 1986; Werling et al., 2010), with convex smoothing and piecewise-jerk optimization making the structure production-ready (Fan et al., 2018; Y. Zhang et al., 2020; Zhou et al., 2021). Temporal optimization, iterative anchoring, candidate pairing, and risk-aware coordination feed information back between stages (Chen, Zhan, & Tomizuka, 2019; Han, Leng, Zhang, & Xiong, 2026; C. Liu, Zhan, & Tomizuka, 2017; F. Liu, Zhao, Su, & Wen, 2024; Z. Zhang et al., 2024), yet arrival-time updates cannot recover a lateral alternative discarded by an irreversible local side decision.

2.3. Safe-corridor and homotopy-corridor methods

Safe-corridor methods make the feasible region explicit before continuous optimization. Spatial flight corridors bridge discrete search and trajectory optimization (S. Liu et al., 2017); driving work extends this to spatiotemporal lane-change or Frenet corridors (Tariq, Yeh, Singh, Isele, & Bae, 2025; Yoon et al., 2024). MIKU shares the explicit-feasible-region principle but targets mutually consistent SL and ST bounds without replacing the path and speed QPs.

2.4. Interaction prediction and safety margins

Prediction-aware planning targets urban regimes where static footprints under-represent pedestrian and vehicle behavior (Jeong & Yi, 2021; B. Yang, Yan, Wang, & Nakano, 2023). Since predictions remain uncertain, MIKU uses nominal arrivals to rank alternatives while bounded position and velocity errors expand the occupancy set for collision checking. Threat-dependent margins allocate lateral space by urgency, overlap, relative motion, object type, and local density; time-to-collision enters as one normalized urgency factor among these terms (Minderhoud & Bovy, 2001).

Table 1 summarizes the solver-level distinction: joint and mixed-integer planners coordinate arrival timing and group-level lateral decisions by replacing or enlarging the two-QP interface, safe-corridor methods construct collision-free sets yet alter solver structures or isolate corridors sequentially, while MIKU’s arrival-conditioned SL projection, threat-aware group partition, and temporal homotopy graph map coordinated decisions directly into box constraints for the downstream path and speed QPs.

Table 1. Planning families grouped by space–time coordination style and solver structure. The rightmost column maps each family to the three constraint-interface gaps identified in Section 1, specifically time-blind occupancy merging, greedy obstacle-wise deadlocks, and cross-stage homotopy loss.

Family	Space–time coordination	Solver implication	Interface-level gap addressed here
Time-blind PVD	Sequential path then speed	Two structured QPs	Loses arrival-conditioned occupancy and group choices
Iterative PVD	Repeated path–speed updates	Multiple solver calls	Cannot restore discarded lateral classes by timing alone
Joint lattice or mixed-integer planning	Coupled state or discrete variables	Expanded search or a different optimizer	Does not preserve the original two-QP interface
Safe-corridor planning	Convex feasible sets before optimization	Corridor generator plus continuous solver	Often focuses on a selected corridor rather than coupled spatial and temporal classes
MIKU	Spatial and temporal homotopies mapped into bounds	Retains the two downstream QPs	Resolves arrival timing, group-level side decisions, and cross-stage constraint transfer

3. Problem formulation

We consider an autonomous vehicle navigating a roadway over planning horizon T_{pred} in the presence of $k(t)$ static and dynamic obstacles. Given vehicle kinematic states and obstacle motion predictions, our objective is to systematically construct the spatial and temporal constraint interface $(\mathcal{B}_l, \mathcal{B}_s)$ for two-stage path–velocity optimization without altering the structure of downstream convex quadratic programs. Following standard formulations in autonomous vehicle motion planning, we focus on behavioral- and trajectory-level optimization; low-level tracking controllers are assumed to track reference trajectories, while communication delays and low-level powertrain dynamics are decoupled from the upper-level trajectory generation.

3.1. Path–velocity decomposition interface

We formulate vehicle motion planning in a road-aligned Frenet coordinate frame defined with respect to a designated smooth reference line. In this frame, the longitudinal coordinate s denotes the arc length along the reference line, representing the vehicle’s forward travel distance, while the lateral coordinate l denotes the signed orthogonal displacement from the reference line.

Under the two-stage path–velocity decomposition framework, the complete spatio-temporal motion planning problem is decoupled into two sequential stages: first, solving for a spatial path profile $l(s)$ that determines lateral maneuvering and obstacle avoidance along the longitudinal station; second, solving for a longitudinal trajectory profile $s(t)$ that governs the vehicle’s forward position, speed, and acceleration over the planning horizon $t \in [0, T_{\text{pred}}]$.

To render these continuous profiles tractable for real-time numerical optimization, the curves are discretized over spatial stations s_j for $j = 0, \dots, K$ and temporal steps t_r for $r = 0, \dots, N$. Let the discrete path state vector be $\mathbf{x}_p = (l_j, l'_j, l''_j)_{j=0}^K$, where $l'_j = dl/ds$ and $l''_j = d^2l/ds^2$ represent the lateral heading deflection and curvature rate, respectively. Let the discrete longitudinal trajectory state vector be $\mathbf{x}_v = (s_r, v_r, a_r)_{r=0}^N$, where $v_r = \dot{s}(t_r)$ and $a_r = \ddot{s}(t_r)$ denote the longitudinal velocity and acceleration. The two planning stages are sequentially solved via convex

quadratic programs:

$$\mathbf{x}_p^* = \arg \min_{\mathbf{x}_p} \frac{1}{2} \mathbf{x}_p^\top H_p \mathbf{x}_p + q_p^\top \mathbf{x}_p, \quad (1)$$

$$\text{s.t. } A_p \mathbf{x}_p = b_p, \quad l_j^- \leq l_j \leq l_j^+, \quad (2)$$

$$\mathbf{x}_v^* = \arg \min_{\mathbf{x}_v} \frac{1}{2} \mathbf{x}_v^\top H_v \mathbf{x}_v + q_v^\top \mathbf{x}_v, \quad (3)$$

$$\begin{aligned} \text{s.t. } A_v \mathbf{x}_v &= b_v, \quad s_r^{lb} \leq s_r \leq s_r^{ub}, \\ 0 &\leq v_r \leq v_{\max}, \quad a_{\min} \leq a_r \leq a_{\max}. \end{aligned} \quad (4)$$

Here, H_p and H_v encode trajectory smoothness costs, while A_p and A_v enforce piecewise-jerk kinematic transition equalities between adjacent discretization points. The operational environment is exposed to the solvers strictly through the spatial path boundaries $\mathcal{B}_l = \{[l_j^-, l_j^+]\}_{j=0}^K$ and temporal position boundaries $\mathcal{B}_s = \{[s_r^{lb}, s_r^{ub}]\}_{r=0}^N$. MIKU reconstructs these constraint interfaces while preserving the objective matrices, kinematic equalities, and convex QP solver formulations.

3.2. Center-coordinate obstacle model

For obstacle i , let $O_i(t) = [s_i^-(t), s_i^+(t)] \times [l_i^-(t), l_i^+(t)]$ denote its physical bounding box. With road limits $[l_{\text{road}}^-, l_{\text{road}}^+]$, vehicle width W_{ego} , and roadside margin d_{road} , the feasible interval for the vehicle center is

$$[L^-, L^+] = [l_{\text{road}}^- + W_{\text{ego}}/2 + d_{\text{road}}, l_{\text{road}}^+ - W_{\text{ego}}/2 - d_{\text{road}}]. \quad (5)$$

The obstacle occupancy interval is expanded once by half the vehicle width and a threat-dependent buffer $d_{\text{buf},i}$:

$$[u_i, v_i] = [l_i^- - W_{\text{ego}}/2 - d_{\text{buf},i}, l_i^+ + W_{\text{ego}}/2 + d_{\text{buf},i}]. \quad (6)$$

All quantities in Equations 5–6 are defined with respect to the vehicle center, ensuring that geometric dimensions are not subtracted repeatedly in downstream modules.

Assigning each of the k active obstacles at a given longitudinal station to the left or right side via decision vector $\mathbf{d}$ yields

$$l^-(\mathbf{d}) = \max \left(L^-, \max_{i \in \mathcal{L}} v_i \right), \quad l^+(\mathbf{d}) = \min \left(L^+, \min_{i \in \mathcal{R}} u_i \right), \quad (7)$$

with empty index sets omitted. The resulting lateral band is feasible when the available clearance satisfies $W(\mathbf{d}) = l^+(\mathbf{d}) - l^-(\mathbf{d}) \geq \varepsilon$ for a minimum width threshold ε .

3.3. Baseline failure mechanisms and evaluation criteria

A time-blind path stage uses the full prediction horizon union

$$\overline{O}_i = \bigcup_{t \in [0, T_{\text{pred}}]} O_i(t), \quad (8)$$

treating spatial occupancies across different time instances as simultaneously blocked. Furthermore, greedy obstacle-wise side decisions can push the upper and lower bounds in Equation 7 past each other, resulting in $l^+(s) < l^-(s)$ and falsely declaring an artificial blockage even when a collision-free spatio-temporal passage exists.

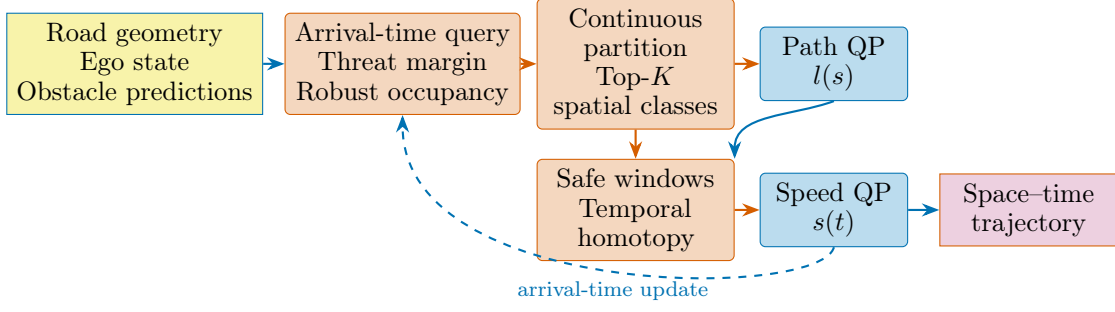


Figure 2. MIKU compiles traffic predictions into convex box bounds $\mathcal{B}_l$ and $\mathcal{B}_s$ for the retained Path QP and Speed QP without changing the solver formulations.

An episode is deemed successful when the optimized trajectory composed of $\mathbf{x}_p^*$ and $\mathbf{x}_v^*$ remains collision-free throughout $[0, T_{\text{pred}}]$ and reaches $s \geq s_{\text{end}} - 1$ m, with the complete operational evaluation metrics detailed in Section 5.

4. MIKU methodology

4.1. Overview

To resolve the spatial blockage and temporal ordering issues identified in baseline PVD planning, MIKU constructs an interaction-aware constraint interface $(\mathcal{B}_l, \mathcal{B}_s)$ between the traffic environment and the downstream optimization solvers. The methodology operates through four coordinated modules: first, arrival-time projection, presented in Section 4.4, which isolates temporally disjoint obstacle occupancies; second, threat-aware safety margins, detailed in Section 4.2, which adaptively scale lateral clearance based on interaction urgency; third, continuous lateral partition, formulated in Section 4.3, which determines group-level side assignments and extracts candidate spatial corridors; and fourth, a temporal homotopy graph, developed in Section 4.5, which constructs collision-free station-time intervals for velocity optimization. As illustrated in Figure 2 and Algorithm 1, these modules systematically enrich the constraint interfaces while strictly preserving the standard convex QP formulations of (2) and (4). Table 4 details the incremental isolation of each component across rows A1–A6.

4.2. Threat-aware margins

Constant margins either close narrow passages or force braking to absorb residual risk. The module tunes only $d_{\text{buf},i}$, leaving the sort and prefix scan of Section 4.3 unchanged; this component is isolated by ablation A4. Urgency, overlap, relative motion, type, and density combine into a threat score

$$\Theta_i = w_1 f_{\text{TTC}}(i) + w_2 f_{\text{overlap}}(i) + w_3 f_{\text{vel}}(i) + w_4 f_{\text{type}}(i) + w_5 f_{\text{inter}}(i). \quad (9)$$

All factors lie in $[0, 1]$, with weights set to 0.30, 0.20, 0.15, 0.10, and 0.25. $\text{TTC}_i = (s_i - s_{\text{ego}}) / (\dot{s}_{\text{ego}} - \dot{s}_i)$ when ego is behind and closing, else $+\infty$; piecewise-linear f_{TTC} is 1 for $\text{TTC}_i \leq 2.0$ s, 0 for ≥ 7.0 s, linear between (Minderhoud & Bovy, 2001). Overlap and relative-speed:

$$f_{\text{overlap}}(i) = \min \left(1, \frac{\max(0, \min(l_i^+, l_{\text{ego}}^+) - \max(l_i^-, l_{\text{ego}}^-))}{\max(0.1, \min(W_i, W_{\text{ego}}))} \right), \quad (10)$$

$$f_{\text{vel}}(i) = \sigma(v_{\text{rel},i}/v_{\text{max}}), \quad \sigma(x) = \frac{1}{1 + e^{-5x}}. \quad (11)$$

with $W_i = l_i^+ - l_i^-$ and the cap keeping $f_{\text{overlap}} \in [0, 1]$. Type factors for vulnerable road users, motor vehicles, unknown movable objects, static obstacles, small markers: 1.0, 0.7, 0.5, 0.3, 0.15; density uses a 10 m neighborhood, zero for a lone obstacle. Extra margin:

$$d_{\text{buf},i} = d_{\text{buf},\min} + (d_{\text{buf},\max} - d_{\text{buf},\min})\Theta_i, \quad (12)$$

with $[d_{\text{buf},\min}, d_{\text{buf},\max}] = [0.10, 0.40]$ m. Operationally, this module widens lateral clearance near vulnerable road users and closing high-speed obstacles while permitting tighter passage around slow or distant static objects, directly trading available corridor width for interaction-dependent safety margin.

4.3. Continuous lateral partition and maximum gap

Obstacle-wise policies can push bounds past one another, emptying the center; MIKU changes only $\mathcal{L}/\mathcal{R}$ assignment in (7). Overlapping longitudinal intervals form components in $O(n \log n)$; within each, sort obstacles by u_i , breaking ties by descending v_i .

Lemma 4.1 (Continuous optimal partition). *For (7), some optimal assignment has obstacles sorted by u_i as a prefix in L and suffix in R ; only $k + 1$ splits need be considered.*

Proof. With $\beta = \min_{i \in \mathcal{R}} u_i$, moving every $i \in \mathcal{L}$ with $u_i \geq \beta$ to $\mathcal{R}$ leaves the upper bound unchanged and cannot raise the lower bound, so width does not decrease. Iterating yields a continuous split; empty sides give $p = 0$ or k ; descending v_i breaks ties. $\square$

For split p , prefix maximum and candidate upper edge:

$$V_0 = L^-, \quad V_p = \max \left(L^-, \max_{1 \leq q \leq p} v_{(q)} \right), \quad (13)$$

$$U_p = \begin{cases} \min(L^+, u_{(p+1)}), & p < k, \\ L^+, & p = k, \end{cases} \quad g_p = U_p - V_p. \quad (14)$$

Theorem 4.2 (Maximum-gap optimality). *For the fixed-section interval model, the optimal center-band width is $W^* = \max_{p=0,\dots,k} g_p$, obtained by sorting and one prefix scan in $O(k \log k)$.*

Proof. Equation 7 gives width g_p at split p ; Lemma 4.1 makes $k + 1$ splits sufficient. As illustrated in Figure 3, for $k = 4$ sorted obstacle intervals, the formulation evaluates $k + 1 = 5$ candidate continuous split boundaries between road bounds L^- and L^+ . Split $p = 2$ achieves the maximum feasible gap $g_2 = U_2 - V_2$, fully eliminating the exponential 2^k combinatorial search without optimality loss under the fixed-section model. $\square$

MIKU keeps $Q = 3$ widest positive bands; each candidate band B_{jq} with center c_{jq} and width g_{jq} enters a dynamic programming optimization that minimizes

$$-\sum_j g_{jq_j} + \lambda_s \left(|c_{1q_1} - l_0| + \sum_{j>1} |c_{jq_j} - c_{j-1,q_{j-1}}| \right), \quad (15)$$

via the recurrence

$$D_{jq} = -g_{jq} + \min_r \{D_{j-1,r} + \lambda_s |c_{jq} - c_{j-1,r}|\}. \quad (16)$$

The weighting parameter λ_s is calibrated in Section 5, and ablation A3 isolates Top- K spatial homotopy.

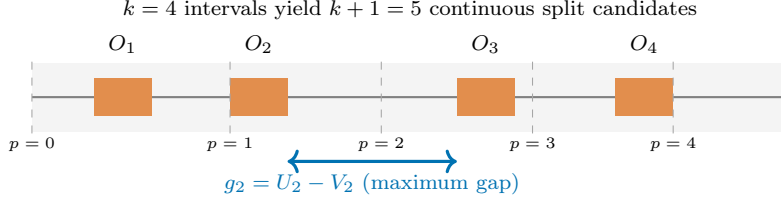


Figure 3. Sorted forbidden intervals $[u_i, v_i]$ yield $k + 1 = 5$ split candidates between L^- and L^+ . Split $p = 2$ attains $W^* = g_2 = U_2 - V_2$ by an $O(k \log k)$ prefix scan (Lemma 4.1, Theorem 4.2).

4.4. Arrival-time-dependent projection

Time-blind projection merges occupancies at different arrival times into one blocking interval; parameterizing each by $\tau(s)$ exposes only obstacles present when ego crosses. Sort, scan, and QP structure are unchanged; ablation A1 isolates this projection, while its closed-loop behavior under rolling replanning is evaluated in Section 6.3. Under constant acceleration,

$$\tau(s) = \frac{-v_{\text{ego}} + \sqrt{v_{\text{ego}}^2 + 2a_{\text{ego}}(s - s_{\text{ego}})}}{a_{\text{ego}}}, \quad (17)$$

with the constant-speed limit for $|a_{\text{ego}}| < 10^{-3}$; a negative discriminant marks stations outside the horizon. In rolling replanning $\tau(s)$ is interpolated from the previous speed solution and refreshed each cycle; sensitivity to this profile is analyzed in Section 5. Scenarios use $s_i(t) = s_{i,0} + v_{s,i}t$, $l_i(t) = l_{i,0} + v_{l,i}t$; bounded prediction errors give

$$\epsilon_{s,i}(t) = \epsilon_{s0,i} + t\epsilon_{vs,i}, \quad \epsilon_{l,i}(t) = \epsilon_{l0,i} + t\epsilon_{vl,i}. \quad (18)$$

Footprints are expanded before the ST map; nominal $\tau(s)$ ranks alternatives, collision checking uses expanded occupancy. Linear-in- t growth provides a nominal occupancy envelope; the parameters $\epsilon_{s0,i}$, $\epsilon_{vs,i}$, $\epsilon_{l0,i}$, and $\epsilon_{vl,i}$ are configured in Section 5 and evaluated under ablation A6.

4.5. Temporal homotopy and ST transfer

Two space-time solutions belong to the same temporal homotopy class if they realize the same passing order relative to each dynamic obstacle. The graph constructed below selects candidate classes over discrete passage-window sequences. Without temporal homotopy, before- and after-obstacle alternatives collapse and causal order is lost. The graph modifies only the ST bounds fed to the speed QP, preserving its convex objective and kinematic equalities in (4). The parameters η_c and ϵ_s are set in Section 5, while ablations A5 and A6 isolate their individual contributions. With $I_{cr} = [t_{cr}^{in}, t_{cr}^{out}]$ each path-tube overlap for conflict c , after expansion

$$\mathcal{W}_c = [0, T] \setminus \bigcup_r [t_{cr}^{in} - \eta_c, t_{cr}^{out} + \eta_c]. \quad (19)$$

Intersecting with earliest reachable arrival gives $W_{cq} = [\alpha_{cq}, \beta_{cq}]$. Station-ordered conflicts connect when

$$\hat{t}_c \geq \hat{t}_{c-1} + \frac{s_c - s_{c-1}}{v_{\max}}, \quad \hat{t}_c \in W_{cq}. \quad (20)$$

Node cost is $\tau(s_c)$ deviation, edge cost penalizes side-switching. A bounded-width DP picks one causal sequence for the speed QP:

$$\begin{aligned} s_j^{ub} &= \min(s_j^{ub,0}, s_c^- - \epsilon_s), & t_j &< \alpha_{cq}, \\ s_j^{lb} &= \max(s_j^{lb,0}, s_c^+ + \epsilon_s), & t_j &\geq \beta_{cq}. \end{aligned} \tag{21}$$

If no sequence is reachable, a stopping bound precedes the first unreachable conflict. From an operational perspective, the temporal homotopy graph preserves the causal order of “pass before” versus “yield after” decisions for each obstacle, ensuring the speed planner receives physically consistent timing constraints rather than conflated or collapsed passage alternatives.

4.6. Algorithm and complexity

For K path samples, m components, Q candidates, temporal beam B , K_s prefixes: sorting/grouping $O(n \log n)$; spatial DP $O(mK_sQ^2 \log(K_sQ))$; temporal beam $O(hBR \log(BR))$ over h conflicts and R windows. Occupancy queries $O(nK)$; naive interaction-density $O(n^2)$. Default parameters are set to $Q = 3$, $K_s = 3$, and $B = 8$ as specified in Section 5. Downstream path and speed QPs maintain complexities of $O(K)$ and $O(N_t)$ per (2) and (4).

5. Experimental design

The experimental evaluation investigates planning performance across three distinct tiers: large-scale open-loop statistical simulation, closed-loop rolling replanning, and production software integration within Apollo Planning. Statistical experiments use deterministic scenario generators and fixed random seeds so that every method receives identical inputs, with collision and clearance evaluated against separately maintained ground-truth trajectories.

5.1. Protocols and scenarios

Each scenario uses a fixed horizon $T \in \{9, 10, 12, 14\}$ s assigned by family: 9 s for crossing pedestrians and delayed crossings; 10 s for vehicle cut-ins, parked-with-oncoming traffic, and prediction noise; 12 s for narrow multi-obstacle situations; and 14 s for interleaved dynamic traffic. An episode times out when the clock reaches T without achieving $s_{\text{end}} \geq s_{\text{max}} - 1$ m. Four disjoint protocols are used.

Open-loop evaluation over 3,500 scenarios. 500 seeds in each of seven families: crossing pedestrians, vehicle cut-ins, parked with oncoming traffic, narrow multi-obstacle, interleaved dynamic, prediction noise, delayed crossings. Parameters are sampled within family ranges. Planning and truth scenarios coincide except in prediction-noise, where bounded errors are injected only into planning input.

Fixed-section geometry over 4,000 instances. 20 deterministic seeds with 200 fixed-section interval instances per seed; each has zero to eight possibly overlapping, nested, or road-crossing forbidden intervals. The scan is compared with exhaustive enumeration of all 2^k side assignments on center-band width.

Rolling-replanning evaluation over 700 scenarios. 100 seeds per family. The ego vehicle is initialized from truth state (s_0, l_0, v_0, a_0) at $t = 0$ with a cold-start first cycle that does not use warm-started QP duals or committed homotopy. Every 0.5 s the first planned segment executes, obstacle states update, and the planner is called on shifted horizon $\max(2.0 \text{ s}, T - t)$ until the vehicle achieves arrival at $s \geq s_{\text{max}} - 1$ m or reaches T .

Finite-grid joint reference over 70 scenarios. Stratified with 10 seeds per family and grid spacings $\Delta t = 0.5$ s, $\Delta s = 0.5$ m, $\Delta l = 0.25$ m, $\Delta v = 1.0$ m/s, beam width 20,000.

Algorithm 1. MIKU constraint generation and mapping

Input: Road bounds $[L^-, L^+]$, ego state $(s_{\text{ego}}, v_{\text{ego}}, a_{\text{ego}})$, predicted trajectories $\{O_i(t)\}_{i=1}^n$
Output: Path bounds $\mathcal{B}_l = \{[l_j^-, l_j^+]\}$; ST bounds $\mathcal{B}_s = \{[s_r^{lb}, s_r^{ub}]\}$

/ Stage 1: Arrival-time estimation and threat-aware intervals */*

- 1 **foreach** station s_j **do**
- 2 Compute $\tau(s_j)$ via (17)
- 3 **foreach** obstacle i with $s_i^-(\tau) \leq s_j \leq s_i^+(\tau)$ **do**
- 4 Compute threat score Θ_i via (9); set $d_{\text{buf},i}$ via (12)
- 5 Expand to center-coordinate interval $[u_i, v_i]$ via (6)

/ Stage 2: Connected-component grouping and continuous lateral partition */*

- 6 Sort obstacle station intervals by start position; form connected components via interval union in $O(n \log n)$
- 7 **foreach** component $\mathcal{C}_m$ with k_m active obstacles **do**
- 8 Sort obstacles in $\mathcal{C}_m$ by $(u_i, -v_i)$
- 9 $V_0 \leftarrow L^-$
- 10 **for** $p \leftarrow 1$ to k_m **do**
- 11 $V_p \leftarrow \max(V_{p-1}, v_{(p)})$ *// prefix maximum of upper edges*
- 12 $U_p \leftarrow \min(L^+, u_{(p+1)})$; $g_p \leftarrow U_p - V_p$
- 13 $g_0 \leftarrow \min(L^+, u_{(1)}) - L^-$; $g_{k_m} \leftarrow L^+ - V_{k_m}$
- 14 Keep Q splits with largest positive g_p ; record centers c_{jq} and widths g_{jq}

/ Stage 3: Spatial homotopy selection via dynamic programming */*

- 15 **for** $j \leftarrow 1$ to K **do**
- 16 **foreach** candidate $q \in \{1, \dots, Q\}$ **do**
- 17 $D_{jq} \leftarrow -g_{jq} + \min_r \{D_{j-1,r} + \lambda_s |c_{jq} - c_{j-1,r}|\}$ *// (16)*
- 18 Backtrack optimal $\{q_j^*\}$; write $l_j^- \leftarrow V_{p_j^*}$, $l_j^+ \leftarrow U_{p_j^*}$ into $\mathcal{B}_l$

/ Stage 4: Temporal homotopy graph and ST bound transfer */*

- 19 **foreach** path-obstacle conflict c at station s_c **do**
- 20 Compute overlap intervals $\{I_{cr}\}$ between planned path and expanded occupancy tube
- 21 $\mathcal{W}_c \leftarrow [0, T] \setminus \bigcup_r [t_{cr}^{\text{in}} - \eta_c, t_{cr}^{\text{out}} + \eta_c]$ *// (19)*
- 22 Intersect $\mathcal{W}_c$ with earliest reachable arrival to obtain windows W_{cq}
- 23 Select causal temporal sequence via bounded-width DP over (20)
- 24 **foreach** selected window $[\alpha_{cq}, \beta_{cq}]$ **do**
- 25 $s_r^{ub} \leftarrow \min(s_r^{ub}, s_c^- - \epsilon_s)$ for $t_r < \alpha_{cq}$
- 26 $s_r^{lb} \leftarrow \max(s_r^{lb}, s_c^+ + \epsilon_s)$ for $t_r \geq \beta_{cq}$ *// (21)*

/ Optional: alternating refinement */*

- 27 **if** refinement enabled **and** iteration < 2 **then**
- 28 Update $\tau(s)$ from speed QP solution; go to Stage 1
- 29 **return** $\mathcal{B}_l, \mathcal{B}_s$

Table 2. Fixed planning and uncertainty parameters used in all reported experiments.

Parameter	Value	Role
$(w_1, \dots, w_5)$	(0.30, 0.20, 0.15, 0.10, 0.25)	threat-score weights
TTC breakpoints	2.0, 7.0 s	urgency normalization
$(\delta_{\min}, \delta_{\max})$	(0.10, 0.40) m	adaptive obstacle buffer
Q, K_s, λ_s	3, 3, 0.05	spatial bands, Top- K , transition weight
B	8	temporal-graph beam width
η_c	$0.10 + 0.05 \min(v_l , 2)$ s	occupancy-window expansion
ϵ_s	0.05 m	longitudinal ST guard
$(\epsilon_{s0}, \epsilon_{l0})$	(0.35, 0.18) m	initial prediction-error bounds
$(\epsilon_{vs}, \epsilon_{vl})$	(0.30, 0.12) m/s	prediction-error growth bounds
Path discretization	0.5 m	path-bound sampling
Speed discretization	0.1 s	speed-QP sampling
Planning iterations	at most 2	initial solve plus refinement

5.2. Baselines and implementation

The matched PVD baselines are defined by information exchanged at the constraint interface:

- B0 is time-blind and uses the fixed local side policy.
- B1 adds arrival-time-dependent projection but retains obstacle-wise greedy side selection.
- B2 allows three arrival-time refinement iterations in the B1 policy.
- MIKU adds group-level continuous partition, Top- K spatial homotopy, threat-aware margins, uncertainty-expanded occupancy, temporal homotopy, and at most two planning iterations with alternating updates.

All four share identical path and speed QP objectives, kinematic constraints, input, and metric implementation. Both piecewise-jerk QPs use OSQP at default tolerances with no per-call wall-clock cap on the 3,500-scenario runs. Baseline B2 uses the same QPs inside three arrival-time refinement iterations. Baseline B3 is a deterministic beam-search dynamic program on the (t, s, l, v) grid with a beam width of 20,000. Its per-transition cost weights speed deviation from v_0 , longitudinal acceleration, lateral offset, and lateral acceleration minus a progress reward; transitions that are kinematically inadmissible or collide with uncertainty-expanded obstacles are pruned before ranking. MIKU hyperparameters were fixed before evaluation and are reported completely in Table 2. Runs used an Intel i9-14900HX with 64 GB RAM. Apollo integration confirms compatibility with Apollo 11.0 Planning.

5.3. Outcomes and statistical analysis

The primary outcome is collision-free goal reaching: a run succeeds when collision-free over the horizon and reaching $s_{\text{end}} \geq s_{\text{max}} - 1$ m. Collision rate uses truth rectangles; degraded-stop rate records runs that stop or fail to reach the goal without collision. Progress ratio, average speed, jerk RMS, minimum signed clearance, minimum longitudinal TTC, penalised travel time, and planning latency give safety, comfort, and computational context. Penalised travel time equals arrival time for collision-free arrivals and family-specific T from Section 5.1 plus 5 s otherwise. Runtime includes the full method call; B2 includes all coordination iterations.

For each fixed seed we compute MIKU minus baseline differences before aggregation. Continuous metrics use 5,000-resample paired percentile bootstrap intervals; binary outcomes use exact McNemar tests; effect sizes are paired Cohen’s d_z . Runtime is reported at the 50th, 95th, and 99th percentiles.

Table 3. Results on 3,500 paired scenarios across seven traffic families. Rates in percent, continuous outcomes as means, latency in ms. Bold marks the best value in each row.

Outcome	B0	B1	B2	MIKU
Collision-free goal reaching (%)	62.69	52.37	62.09	77.46
Collision (%)	2.17	2.14	2.23	1.14
Degraded stop (%)	35.40	45.77	35.97	21.51
Progress ratio (%)	77.52	71.07	76.62	86.45
Jerk RMS (m/s ³)	1.252	1.315	1.300	1.132
Runtime P50 (ms)	10.11	11.37	31.20	9.98
Runtime P95 (ms)	56.79	63.24	169.49	55.93
Runtime P99 (ms)	138.88	169.73	463.05	221.80

Notes: Evaluated across $N = 3,500$ paired simulation scenarios. Rates are reported in percent (%), continuous outcomes as sample means, and planning latency in milliseconds (ms). P50, P95, and P99 represent the 50th, 95th, and 99th percentiles of complete-planning runtime. Bold indicates the best performing method in each row.

6. Results

6.1. Comparative Benchmark Evaluation

In theorem verification, the $O(k \log k)$ scan and exhaustive enumeration agreed on all 4,000 instances, with maximum center-band-width difference below 10^{-12} .

For the primary comparative benchmark, Table 3 reports the 3,500 paired scenarios. MIKU raised collision-free goal reaching from 62.69% under baseline B0 to 77.46%, achieving a paired difference of 14.77 percentage points with a 95% bootstrap confidence interval of 13.54 to 15.97 and an exact McNemar test result of $p = 1.94 \times 10^{-133}$. The collision rate fell from 2.17% to 1.14%, corresponding to a reduction of 1.03 percentage points with a 95% confidence interval from -1.37 to -0.71 and $p = 2.91 \times 10^{-11}$. Mean progress rose by 8.93 percentage points, jerk RMS fell by 0.120 m/s³, and penalized travel time decreased by 0.81 s. The aggregate mean-clearance difference was -0.010 m with a 95% confidence interval of $[-0.024, 0.004]$ m.

Examining family-level effects, Panel (a) of Figure 4 locates the difference: the largest success gain occurs in narrow multi-obstacle traffic with an increase of 65.4 percentage points, followed by delayed crossing with an increase of 26.8 percentage points. Crossing-pedestrian and vehicle cut-in families show no aggregate difference because their geometry admits a single dominant passage on either side; MIKU’s group-level partition and temporal homotopy do not alter the constraint interface in these single-corridor configurations, confirming that the method introduces no regressions where the baseline interface is already adequate. Prediction noise yields a 3.2-point success gain and 7.0-point collision reduction; however, progress drops 1.12 points and jerk RMS rises 0.041 m/s³, an expected side effect of the conservative uncertainty-expanded occupancy tube when the error bound is large. Panel d shows that the paired gain comes from converting degraded stops into successful arrivals.

In comparison against the joint spatiotemporal search reference B3 across 70 finite-grid cases, B3 reached 80.0% and MIKU reached 77.1%, yielding a paired difference of -2.86 percentage points with a 95% confidence interval of $[-11.43, 5.71]$ and $p = 0.754$; median runtime was 1,835.43 ms for B3 versus 10.44 ms for MIKU.

6.2. Component Ablation and Mechanism Attribution

Subtractive ablations on the same 3,500 scenarios use the 5,000-resample bootstrap from Section 5. Removing Top- K spatial homotopy cut success by 10.17 percentage points; removing threat-aware margin, 10.74 points. Removing the temporal homotopy graph cut delayed-crossing success from 89.6% to 66.8%, which represents a 22.8-point loss in that family. Removing the robust occupancy tube raised prediction-noise collision from 3.6% to 11.0%. Arrival projection and longitudinal grouping produce smaller aggregate changes with value concentrated in specific configurations. Panel c of Figure 4 summarizes the tabulated effects.

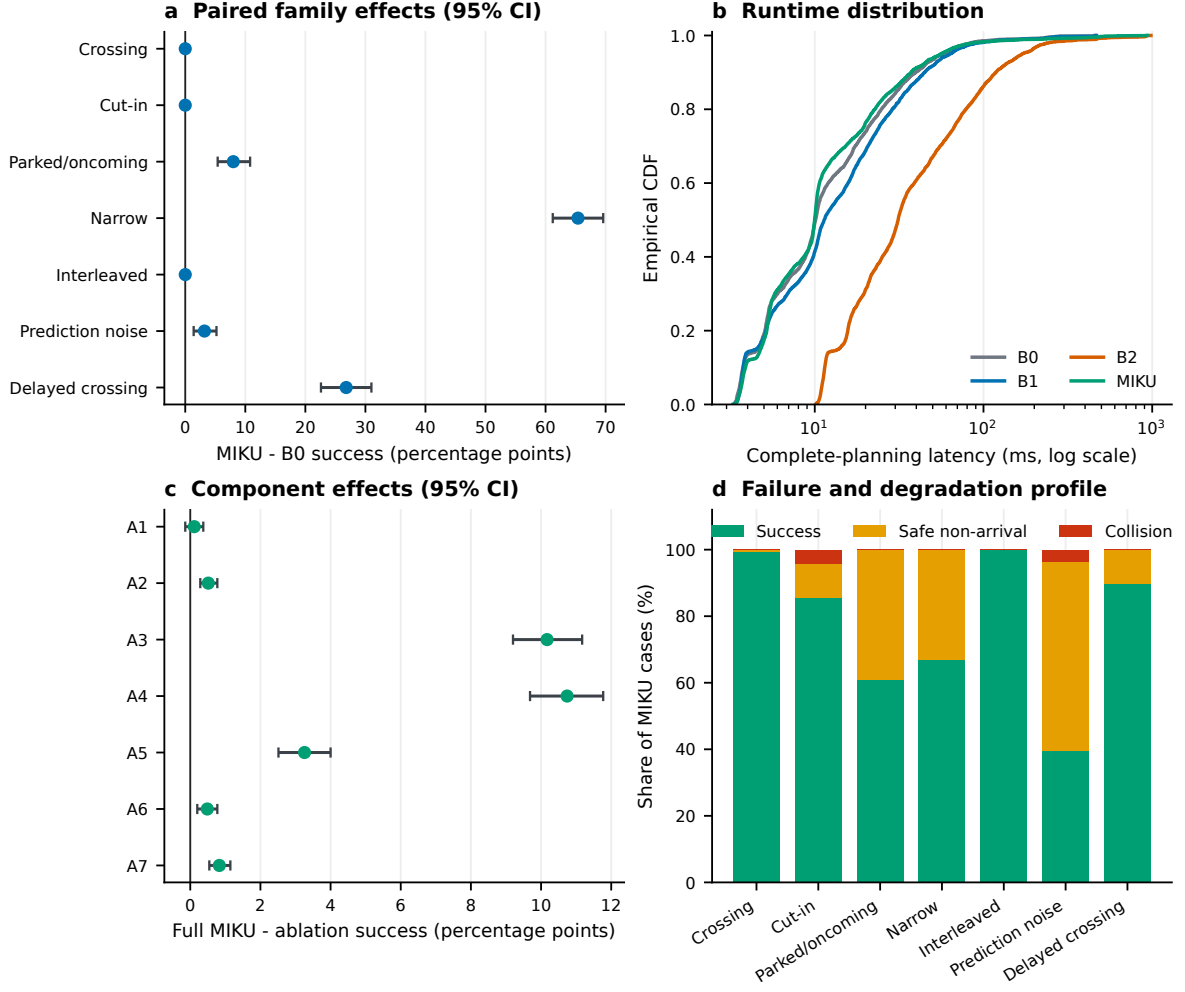


Figure 4. Summary of the 3,500-case paired protocol. Panels show family-wise success differences (a), planning-latency distribution (b), ablation drops (c), and outcome mix (d).

Table 4. Subtractive ablations on the same 3,500 paired scenarios, where Δ columns give signed change in percentage points relative to the full MIKU pipeline. Bold marks the full-method row.

Variant	Removed component	Success (%)	Δ Succ.	Collision (%)	Δ Coll.	Progress (%)	Δ Prog.
A1	Arrival-time projection	77.34	-0.12	1.23	+0.09	86.73	+0.28
A2	Longitudinal grouping	76.94	-0.52	1.14	0.00	85.89	-0.56
A3	Top- K spatial homotopy	67.29	-10.17	1.14	0.00	79.53	-6.92
A4	Threat-aware margin	66.71	-10.74	1.14	0.00	78.85	-7.60
A5	Temporal homotopy graph	74.20	-3.26	1.17	+0.03	84.29	-2.16
A6	Robust occupancy tube	76.97	-0.49	2.20	+1.06	86.60	+0.15
A7	Alternating refinement	76.63	-0.83	1.03	-0.11	85.92	-0.53
MIKU	None	77.46	—	1.14	—	86.45	—

Notes: Evaluated on the same $N = 3,500$ scenarios using 5,000 bootstrap resamples. Δ denotes the signed change in percentage points relative to the full MIKU pipeline; negative Δ in success or progress and positive Δ in collision represent performance degradation upon component removal. Bold marks the full proposed framework.

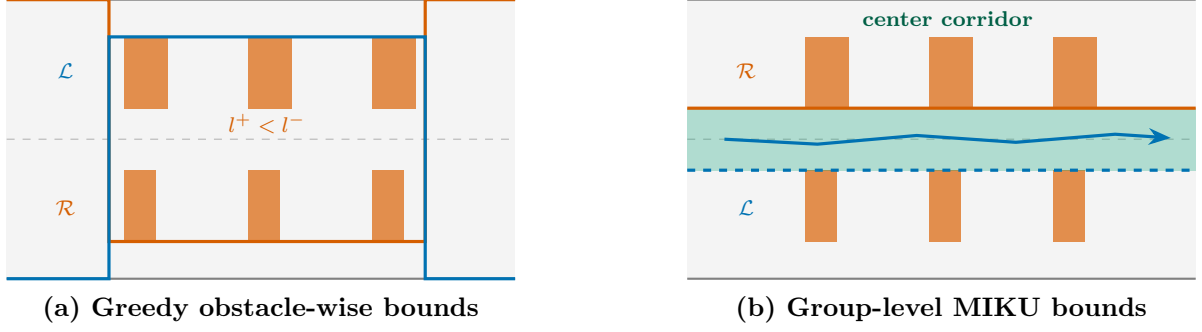


Figure 5. Greedy obstacle-wise bounds empty the feasible interval at the constriction (a). MIKU’s group-level partition keeps a continuous center corridor (b).

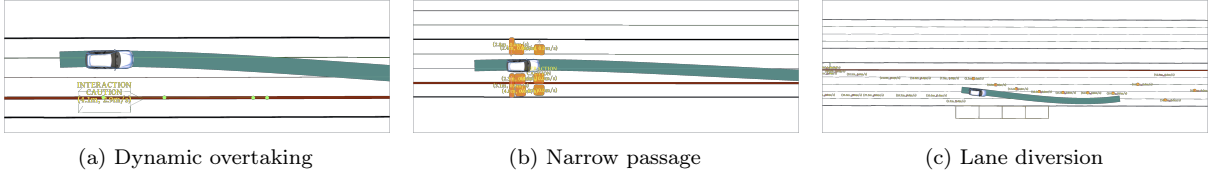


Figure 6. Apollo 11.0 Dreamview episodes of dynamic overtaking (a), narrow passage (b), and lane diversion (c).

Geometrically, as illustrated in Figure 5, in the narrow passage the obstacle-wise policy raises and lowers opposing path bounds until the center interval empties, whereas the group-level partition retains one continuous band once obstacle-side choices are coordinated. In delayed crossing, the temporal graph preserves before- and after-obstacle alternatives and enforces causal order.

6.3. Robustness, Latency Distribution, and Rolling Replanning

In the prediction-noise subgroup, alongside the 3.2-point success gain and 7.0-point collision reduction from Section 6.1, MIKU’s mean progress is 59.73% versus 60.85% and jerk RMS 2.274 versus 2.234 m/s³, consistent with uncertainty-expanded occupancy being conservative when the error bound is large.

Complete-call runtime median and P95 match or beat B0, but the P99 tail does not: MIKU’s P99 is 221.80 ms versus B0’s 138.88 ms, with upper-tail divergence in Panel b of Figure 4. B2 is slower at all quantiles because it always runs three coordination iterations.

The rolling-replanning protocol evaluates cumulative closed-loop behavior. Across 700 cases, success rises from 62.29% to 72.29%, representing an improvement of 10.0 percentage points with a 95% confidence interval of 7.57 to 12.57, while collision stays at 1.29% for both methods, mean progress rises from 82.55% to 90.25%, and episode-runtime P95 rises from 827.20 to 1,361.51 ms.

6.4. Production Software-Interface Integration

The Apollo 11.0 integration feeds generated path and ST constraints directly into the production-grade path-boundary and ST-boundary modules, preserving downstream QP objective functions, kinematic equalities, and convex structures. As illustrated in Figure 6, three representative Dreamview scenarios spanning dynamic overtaking, narrow-passage traversal, and single-lane diversion validate smooth trajectory execution and consistent obstacle negotiation within the standard 0.5 s Apollo planning cycle, confirming practical compatibility with deployed autonomous driving platforms.

7. Discussion

7.1. Physical interpretation of performance gains

The 3,500-case paired comparison directly connects theoretical failure mechanisms to concrete quantitative improvements. MIKU raises collision-free goal reaching from 62.69% to 77.46% against baseline B0, yielding a paired increase of 14.77 pp with a 95% confidence interval of 13.54 to 15.97. This performance gain is concentrated in configurations governed by severe spatial or temporal bottlenecks: under narrow multi-obstacle geometry, where greedy obstacle-wise decisions cross each other and collapse opposing path bounds such that $l^+ < l^-$, MIKU’s group-level partition restores a continuous, feasible center corridor, increasing success by 65.4 pp; under delayed-crossing geometry, where time-blind bounds falsely treat future arrival-time occupancies as immediate blockages, arrival projection and temporal homotopy preserve reachable passing windows before and after the crossing obstacle, increasing success by 26.8 pp. In contrast, in scenarios naturally characterized by a single dominant corridor, such as simple pedestrian crossings and high-speed vehicle cut-ins, performance remains consistent with the baseline, verifying that MIKU specifically targets and resolves diagnosed interface bottlenecks.

Ablation experiments substantiate this mechanism attribution. Removing Top- K spatial homotopy under ablation A3 or the threat-aware margin under ablation A4 reduces success by 10.17 pp and 10.74 pp, respectively, demonstrating that group-level lateral coordination provides the foundational clearance necessary for spatial feasibility. Removing the temporal homotopy graph under ablation A5 selectively cuts delayed-crossing success from 89.6% to 66.8%, a loss of 22.8 pp that confirms its decisive role when multiple temporal passages coexist. Finally, removing the robust occupancy tube under ablation A6 causes prediction-noise collision rates to jump from 3.6% to 11.0%, affirming that uncertainty-aware buffer expansion is essential for absorbing real-world tracking and perception noise (F. Liu et al., 2024; B. Yang et al., 2023; Yoon et al., 2024).

These findings align with prior decomposition-based planning literature while clarifying the locus of improvement. Fan et al. (2018) and Y. Zhang et al. (2020) demonstrated that piecewise-jerk QP formulations achieve production-grade smoothness and latency. Corridor-based approaches (S. Liu et al., 2017; Yoon et al., 2024) showed that explicit feasible-region construction improves trajectory quality in structured environments; MIKU extends this principle by coupling spatial and temporal corridor selection through a shared homotopy graph. The iterative anchoring strategy of Chen et al. (2019) feeds timing information back between stages, yet our B2 results, reaching 62.09% success and remaining comparable to the time-blind B0, confirm that iterative arrival-time updates alone cannot recover lateral classes discarded by an irreversible initial side decision. For traffic system designers, the practical consequence is that upgrading the constraint interface layer—without replacing the downstream QP solvers—can recover a substantial fraction of the performance gap between decomposed and joint planners while preserving the computational budget required for real-time deployment.

7.2. Computational efficiency and engineering trade-offs

In operational autonomous driving systems, planning algorithms must operate within strict onboard latency budgets, typically 50 to 100 ms. Across 3,500 evaluated instances, MIKU achieves a median runtime of 9.98 ms and a P95 of 55.93 ms, demonstrating that interaction-aware constraint reconstruction adds negligible overhead during nominal driving conditions.

Computational cost concentrates primarily in the distribution tail: complete-call P99 latency rises to 221.80 ms compared with 138.88 ms for B0. This tail behavior reflects the physical reality of dense multi-obstacle traffic: branch expansion in the spatial partition and temporal homotopy graph only activates when multiple obstacles simultaneously compete for shared roadway space. For production deployment, timing guards should be budgeted against the P99 tail rather than median latency. Furthermore, under prediction noise, the 7.0 pp collision reduction incurs a modest 1.12 pp progress reduction and a 0.041 m/s³ jerk increase, illustrating an intentional

and controllable engineering trade-off between absolute safety margins and aggressive forward progress.

7.3. Comparison with joint search and iterative baselines

Under the 70-case finite-grid protocol, the joint spatiotemporal reference B3 achieves 80.0% success compared with 77.1% for MIKU, with a paired difference of -2.86 pp, a 95% confidence interval of $[-11.43, 5.71]$, and $p = 0.754$. B3’s median runtime is 1,835.43 ms compared with MIKU’s 10.44 ms, so the finite-grid search trades a modest success increment for a much higher computational cost.

Conversely, the iterative PVD baseline B2 executes three sequential path–speed iterations, requiring a median runtime of 31.20 ms, yet achieves only 62.09% success, which is comparable to B0’s 62.69% and 15.37 pp lower than MIKU.

7.4. Comprehensive insights from multi-protocol evaluation

The multi-tiered experimental framework evaluates planning performance across distinct operational dimensions. The open-loop 3,500-scenario protocol provides statistical power and mechanism isolation under controlled environmental distributions. The 700-scenario closed-loop rolling replanning protocol demonstrates that open-loop advantages reliably accumulate over extended operational horizons, lifting overall goal-reaching success from 62.29% to 72.29%, an improvement of 10.0 pp, while maintaining trajectory stability across successive planning cycles.

Finally, integration into Apollo 11.0 validates architectural compatibility on an established production platform. Across dynamic overtaking, narrow passage navigation, and lane diversion scenarios in Dreamview, MIKU successfully drives the vehicle without modifying existing quadratic programming solvers or control interfaces.

8. Limitations and future research

While MIKU improves interaction-aware constraint generation in multi-obstacle environments, several boundaries limit the current validity and define concrete next steps.

First, regarding prediction models and road geometry, the present empirical evaluation relies on constant-velocity predictions with bounded noise envelopes along structured roadways. Real-world traffic involves irregular road curvature, topological intersections, and multi-modal behavioral uncertainties from surrounding human drivers. This boundary may overestimate collision-rate reductions when prediction errors become systematic rather than bounded. Future extensions will incorporate calibrated multi-modal trajectory prediction distributions, such as Trajectron++ or MTR models trained on naturalistic driving datasets, and generalize the interval abstraction to curved roads exceeding 0.05 rad/m curvature.

Second, regarding the validation tier, the current study demonstrates algorithmic effectiveness through large-scale numerical simulation and software-in-the-loop tests within Apollo 11.0. Future work will add hardware-in-the-loop latency, sensor-pipeline delay, and chassis dynamics under adverse weather or degraded pavement, together with closed-track experiments and completeness analyses under changing passing-order permutations.

9. Conclusion

This paper presented MIKU, an interaction-aware constraint reconstruction framework for two-stage path–velocity motion planning in dense, dynamic multi-obstacle traffic. By introducing arrival-time-dependent projection, continuous group-level lateral partitioning, threat-aware safety margins, and a temporal homotopy graph, MIKU systematically restores missing interaction

timing and spatial topology to the environmental constraint interfaces while strictly preserving the standard convex quadratic programming formulations of downstream path and speed solvers. Furthermore, the fixed-section maximum-gap theorem guarantees that optimal lateral side assignments can be identified via an $O(k \log k)$ scan, avoiding the exponential combinatorial explosion of traditional multi-obstacle allocation.

Extensive comparative evaluations across 3,500 paired scenarios demonstrated that MIKU improves collision-free goal reaching from 62.7% to 77.5%, representing an increase of 14.8 percentage points, and reduces collision rates from 2.2% to 1.1% relative to conventional time-blind baselines. The performance gains are particularly prominent in challenging narrow passages with a 65.4 percentage point gain and delayed-crossing geometries with a 26.8 percentage point gain, directly confirming that the reconstructed constraints eliminate artificial deadlocks. Ablation studies verified that group-level lateral coordination and threat-aware margins establish viable spatial corridors, while temporal homotopy graphs resolve critical causal timing. With a median complete-planning runtime of 9.98 ms, MIKU satisfies real-time execution constraints in typical autonomous driving control loops.

Software-in-the-loop integration within Apollo 11.0 demonstrated software-interface compatibility with production autonomous driving architectures across representative dynamic overtaking, narrow passage, and lane diversion scenarios without altering native QP solver formulations. Closed-loop rolling replanning further verified trajectory consistency and multi-cycle robustness under the simulation conditions tested. Future work will extend the framework to calibrated multi-modal trajectory predictions, complex curved road networks, and full-scale vehicle-in-the-loop experimental platforms.

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Disclosure of interest

The authors report there are no competing interests to declare.

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